W1-Quiz1 Activity: Introduction to Algorithms

A. Discuss Briefly

1. What is an algorithm, and how is it related to the tasks you perform in your daily life?

- An **algorithm** is a step-by-step set of instructions used to solve a problem or complete a task. In daily life, we use algorithms when following instructions, such as cooking a meal, getting ready for school, or solving a math problem.

2. Why is it important for an IT student to learn algorithms?

- Learning algorithms is important for IT students because it helps them **solve problems logically and efficiently**. Algorithms are also used when creating programs, software, websites, and applications.

B. Step-by-Step Algorithms

1.

1. Start with i = 1.
2. Check if i is less than 10.
3. If true, display the value of i.
4. Multiply i by 2.
5. Check again if i is less than 10.
6. Repeat the process until i is no longer less than 10.
7. Stop the loop.

2.

1. Start with i = 0.
2. Check if i is less than 10.
3. If true, display the value of i.
4. Increase i by 1.
5. Check again if i is less than 10.
6. Repeat until i reaches 10.
7. Stop the loop.

C. Observations

1. What did you observe about the sequence of numbers produced by the first code?
- The numbers are 1, 2, 4, and 8. The number doubles every time.

2. What did you observe about the sequence of numbers produced by the second code?
- The numbers go from 0 to 9 in order. It increases by 1 each time.

3. What is the main difference between the way the two loops progress?
- The first loop doubles the number, while the second loop adds 1 to the number.

4. How does the way an algorithm performs its steps affect its output?
- It affects the numbers or results that the program produces. Different steps can give different outputs.